

Ridge Maged Ibrahim

Software Engineer

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PROFILE SUMMARY

Recently graduated Computer Science (Software Engineering) student with hands-on experience across the full development lifecycle — backend systems, cloud infrastructure, mobile apps, and applied machine learning. Built and shipped real systems spanning Django/Flask backends, a Flutter mobile app, cloud infrastructure on Huawei Cloud and Azure, and ML pipelines that run in real time, not just in a notebook — across 3 internships and a graduation project. Comfortable owning a problem end to end: from architecture through implementation to deployment. Graduated June 2026. Fluent in Arabic (Native) and English (Professional).

EDUCATION

King Salman International University

B.Sc. Computer Science — Software Engineering

Graduated: June 2026

El Nasr Boys' School

Secondary Education | Graduated 2021

MILITARY STATUS

Exemption Approved (Official Certificate Pending)

SKILLS

Languages: Dart, Python, JavaScript, Java, C++, HTML5, CSS

Frameworks: Django, Flask, REST API, Flutter, React, Node.js

Cloud & Databases: Firebase Realtime DB, PostgreSQL, MongoDB, Microsoft Azure, AWS, SQL, NoSQL, SQLite

Tools: Git, GitHub, Docker, Excel (Formulas, Pivot Tables, Charts), IoT, Embedded Systems, Postman, Manual Testing, Test Case Design

AI / ML: Machine Learning, Random Forest, NLP, Data Mining, Scikit-learn, Pandas, NumPy, Embedded ML, Generative AI, LLM APIs

EXPERIENCE

Information Technology Institute (ITI) | Aug 2025 – Sep 2025

Full-Stack Development Intern | Remote

- Built REST APIs and full-stack web applications using Django, Flask, and PostgreSQL, reducing API response time by 30% and enabling real-time features.
- Delivered a full-stack Restaurant Management System as capstone project — schema design (PostgreSQL), backend logic (Django), and frontend integration (JavaScript) — with role-based access across 3 user roles and real-time order tracking.

Huawei ICT Academy | Sep 2024 – Oct 2024

Cloud Computing Intern | On-site

- Provisioned and managed cloud infrastructure on Huawei Cloud using IaaS and PaaS services; led an architecture proposal ranked 1st among cohort peers, achieving a 20% efficiency gain.

Huawei ICT Academy | Jul 2025 – Aug 2025

AI / Machine Learning Intern | On-site

- Developed and evaluated supervised ML models (Random Forest, SVM, K-Means) for classification tasks, improving model accuracy by up to 15% through feature engineering and hyperparameter tuning.
- Gained hands-on exposure across the AI/ML/DL spectrum — neural network fundamentals, deep learning concepts, and model evaluation — applied to real classification and prediction problems.

PROJECTS

IoT Real-Time Health Monitor | 2025 – 2026

Graduation Project | ESP32, Flutter, Python, Firebase Realtime DB, Machine Learning, REST API | GitHub

- Owned the end-to-end design and build of a wearable IoT health monitoring system: architected the data flow from an ESP32 microcontroller and 3 onboard sensors (MAX30102, MPU6050, NTC thermistor) through a REST API into Firebase Realtime DB, and a 4-screen cross-platform Flutter client (Live Data, Analytics Dashboard, Intelligence, Compare Metrics) for iOS and Android. Cut database write volume by ~98% by redesigning the ingestion logic around server-side rolling aggregation.
- Designed and implemented a real-time inference layer combining a Python rule engine and Random Forest ML model to evaluate incoming biometric data and generate health alerts across 4 severity tiers (info / caution / warning / danger), including context-aware logic for sleep mode and activity level.
- Debugged and optimized system performance across the full stack — sensor sampling, API response times, database load, and frontend rendering — under continuous real-time load.

Restaurant Management System | 2025

Django, PostgreSQL, JavaScript, REST API | GitHub

- Developed a full-stack restaurant management platform with role-based access control across 3 user types (admin, staff, kitchen); real-time order tracking and sub-200ms API responses streamlined order-to-kitchen communication and reduced manual coordination overhead.

Arabic NLP Sentiment Analysis | 2026

Python, Machine Learning, NLP, Scikit-learn | GitHub

- Built an Arabic sentiment analysis model using supervised machine learning (Scikit-learn) with a full NLP preprocessing pipeline — Arabic tokenization, stop-word removal, and TF-IDF vectorization — achieving 85%+ classification accuracy on positive/negative customer reviews.
- Processed and analyzed over 66,000 Arabic reviews through the complete ML pipeline — data cleaning, feature extraction, model training, and evaluation — demonstrating scalable text classification on a real-world dataset.

AI Document Processing Assistant | 2026

Personal Project | Python, Grok API, Generative AI

- Built a document processing assistant using the Grok API to automatically summarize, extract key information from, and classify text documents, exploring practical Generative AI use cases beyond traditional ML.

CERTIFICATIONS

- Introduction to IoT — Cisco | 2026
- Introduction to Modern AI — Cisco | 2026
- Data Mining Methods — University of Colorado Boulder | Coursera | 2026
- Advanced Topics in Database Technologies — University of Colorado Boulder | Coursera | 2026
- The Full Stack — Meta | Coursera | 2026
- Natural Language Processing in TensorFlow — DeepLearning.AI | Coursera | 2026
- Introduction to Cloud Computing — IBM | 2025
- Full-Stack Web Development Track — ITI | 2025
- Applied Machine Learning in Python — University of Michigan | Coursera | 2025
- Introduction to Software Testing — University of Minnesota | Coursera | 2025
- Project Management: The Basics for Success — UC Irvine | Coursera | 2025
- Critical Thinking Skills for the Professional — UC Davis | Coursera | 2025
- Cloud Computing — Huawei ICT Academy | 2024
- Artificial Intelligence — Huawei ICT Academy | 2024